

Database Requirements

Library Management System — EECS447: Database Systems

Fall 2026 · Professor Hossein Saiedian

Team Name: DataBASED

1. Introduction

Project Overview:

This database will be a reliable library management system that efficiently manages library books, magazines, digital media, client memberships, borrowing and returning transactions, reservations, and fees. The system will help library staff manage library operations and allow clients to search for items, reserve books, and check their loan status on their library membership. The database will also provide useful reports for both users for library fines, book availability, client activity, and more.

Scope:

The library management system will manage books, digital media, and magazines; client accounts and membership types; borrowing and return transactions; reservations; and notifications for due dates, overdue items, and available reservations. Membership types will determine borrowing limits and fee rules. The system will generate reports on fines, item availability, borrowing trends, and client activity. It will include a staff interface for managing items, loans, returns, and accounts, and a client interface for searching the catalog, reserving items, and checking loan status. Work outside these library collection, circulation, and account functions is beyond the scope of this project.

2. Stakeholders

The main stakeholders would be library members, librarians, library administrators, library managers and database administrators. Library members need to check availability of books and check the books out. Librarians need to be able manage members, find and catalogue books and check books out to members. Library administrators need to be able to manage the database, look at overdue books and make reports. Database administrators need to be able to view, edit, backup and maintain the database to make any changes or fixes.

3. Requirements

Functional Requirements:

- User Administration
 - Create, update, and deactivate client accounts: unique ID, name, contact info, membership type, account status.
 - Support multiple client membership types (Regular, Student, Senior, etc.), each with its own borrowing limit and fee rate.
- Data Entry
 - Allow staff to add new items like **Books**, **Digital Media**, **Magazines**, with their respective attributes.
- Retrieval
 - Allow users to **search the item catalog** by title, author, ISBN, publication year, genre, availability status, issue number, publication date, and other attributes.
 - Allow staff to **search client info/attributes** by their client ID.
- Updates
 - Update **item availability status** when items are borrowed or returned at the library.
 - Update **client status/fines** as fees accumulate or are paid by the client.
- Deletions
 - Allow staff to remove items or client accounts from the system.
- Borrowing and Returns
 - Record borrowing transactions including the item borrowed, client, date, and due date.
 - **Enforce borrowing** limits per client membership type.
 - **Restrict borrowing** of certain items based on the client's membership.
- Reservations and Notifications
 - Allow clients to reserve an item currently on loan.
 - Notify clients of upcoming due dates, overdue items, and reserved items becoming available.
- Report generation
 - Fine calculation report per client
 - Book/item availability report by genre
 - Overdue items report listing all overdue items, the client responsible, and the calculated late fees.
 - Client activity report: client borrowing report, exceeded limits.
 - Monthly summary report: report summarizing the total number of items loaned, total fees collected, and most popular items for the month.

Data Entities:

- Book
 - Item ID
 - Title — name of the book
 - Author — writer
 - Genre — category
 - Publication Year — year published
 - Availability Status — available/checked out
 - Restricted — Restriction level for certain memberships
- Digital Media
 - Item ID
 - Title — name of the item
 - Creator — author/artist/publisher
 - Media Type — eBook, audiobook, DVD, etc.
 - Genre — category
 - Publication Year — year released
 - Availability Status — available/checked out
 - Restricted — Restriction level for certain memberships
- Magazine
 - Item ID
 - Title — magazine name
 - Issue Number — specific issue identifier
 - Publication Date — date of that issue
 - Availability Status — available/checked out
 - Restricted — Restriction level for certain memberships
- User (Client)
 - First Name
 - Last Name
 - Client ID
 - Contact Info — email/phone/address
 - Membership Type
 - Account Status — active/suspended/expired
 - Total Fees Owed — running balance
- Membership Type
 - Type Name — Regular, Student, Senior
 - Borrowing Limit — max items allowed at once
 - Daily Fine Rate — late fee per day
 - Loan Period — allowed days per checkout
- Loan
 - Loan ID
 - Client ID — who borrowed
 - Item ID — what was borrowed
 - Checkout Date — when it was borrowed
 - Due Date — when it should return
 - Return Date — when it was actually returned
 - Fine Amount — calculated late fee

- Reservation
 - Reservation ID
 - Client ID — who reserved
 - Item ID — what's reserved
 - Reservation Date — when the reservation was placed
 - Reservation Status — pending/ready/expired/fulfilled

4. Hardware and Software Requirements

The database will be implemented using MySQL, installed locally on each team member's laptop. The team will use a shared GitHub repository to add features and make changes to the database via SQL scripts. Each member can then pull from the latest branch to run those scripts locally and maintain an identical, up-to-date version of the database.